GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

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| Project | FEED for Gas Gathering Pipelines & Associated Infrastructure |
| Client | Brass Fertilizer and Petrochemical Company Limited |
| Originating Company | Epic Atlantic Limited |
| Document Title | **Constructability Study Report** |
| Document Number | EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-003 |
| Document Revision | R02 |
| Document Status | Issued for Review |
| Originator | Malo Ogbedu |
| Security Classification | Restricted |
| Issue Date | 22-Sep-22 |

# CONSTRUCTABILITY STUDY REPORT

| Rev # | Issue Date | Status Description | Originator | Reviewer | Approver (Epic) | Approver (BFPCL) |
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| R02 | 22.09.22 | Re-issued for Review | M. Ogbedu | M. Ekpunobi | A. Duru | |
| R01 | 14.07.22 | Issued for Review | M. Ogbedu | M. Ekpunobi | A. Duru | |

Revision History

| Rev No. | Date | Description of Revision |
|---|---|---|
| R01 | 14.07.2022 | Issued for Review |
| R02 | 22.09.22 | Re-issued for Review after incorporating TCE comments |
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Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
*for the subsequent project phases, post FEED, it is expected that:*
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

## SUMMARY PAGE

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|---|---|---|
| Client: | Brass Fertilizer & Petrochemical Company Limited | |
| Project: | Gas Processing & Methanol Complex | **Project No.** 11998A |
| Facility: | Gas Gathering Pipelines & Associated Infrastructure | **Doc No.** EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-003 |

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

| REFERENCES | |
|---|---|
| Document No. | Description |
| EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001 | Upstream Gas Gathering Concept Select & Revalidation Report |
| EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001 | Preliminary Simulation – Based on Desktop Route Survey |
| EA/SPDC-BFPC UGS-GEP/CSM-RPT-002 | Gas Gathering & Evacuation Pipelines Concept Screening |
| EA/SPDC-BFPC UGS-GEP/CSM-RPT-001 | Conceptual Study Memorandum |
| EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 | Basis of Design |
| | FISAS Draft ESIA Report (November 2020) |
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## Abbreviations (Including Abbreviations Used in the Constructability Study Worksheet)

| Abbreviation | Meaning |
|---|---|
| ALARP | As low As Reasonable Possible |
| CSR | Corporate Social Responsibility |
| DED | Detail Engineering Design |
| EPC | Engineering Procurement and Construction |
| FEED | Front-End Engineering Design |
| FOC | Fibre Optic Cable |
| FTO | Freedom To Operate |
| GMOU | Global Memorandum Of Understanding |
| HDD | Horizontal Directional Drilling |
| HSE | Health, Safety and Environment |
| IDEC | Import Duty Exemption Certificate |
| ITT | Invitation To Tender |
| LIRA | Logistics Infrastructure and Resources Assessment |
| NAOC | Nigerian Agip Oil Company |
| NDT | Non-Destructive Testing |
| OR & A | Operations Readiness & Assurance |
| PMT | Project Management Team |
| ROW | Right of Way |
| SCD | Sustainable Community Development |
| SPDC | Shell Petroleum Development Company |

# 1. Introduction

## 1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

Gas Gathering Concept, Phase-1 (BBOU & ELEP): E-MF at ELEP and S-MF at BBOU

Nodal Gas Gathering at BBOU & ELEP with BBOU Node Gas Joining ELEP Node for Onward Transportation to the Gas Processing Plant at AKAB

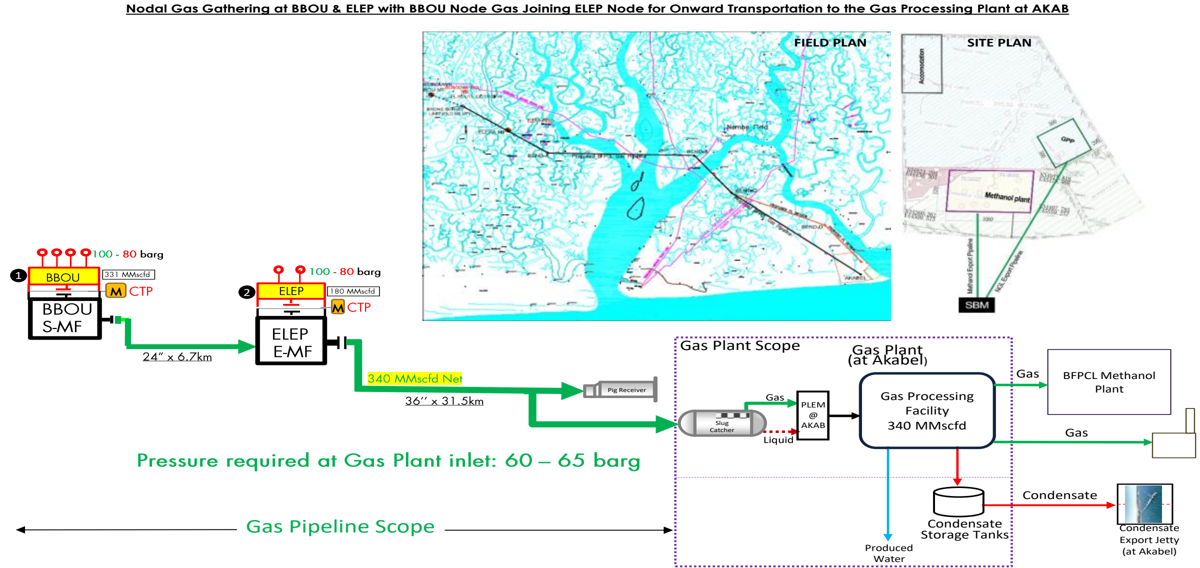


Figure 1: Gas Gathering Concept, Phase 1

## 1.2. Purpose

The purpose of this document is to present the outcome of a constructability workshop that was carried out to identify opportunities and risks facing the project in terms of construction, and to provide recommendations and guidance for installation works whilst recognizing the need to work in accordance with BFPCL HSE and Community Development requirements.

The FEED shall fully comply with all specified relevant contractual requirements.

# 2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

# 3. Scope

The study identified challenges to construction and installation of equipment on the project. Basic construction parameters and criteria were also developed and used as guidelines for evaluating provisions to be made against the identified challenges.

The various equipment and extent of work considered in the review are:

- Manifold at BBOU
- manifold at ELEP
- Pipeline end facilities at AKAB
- Pipelines
- Corrosion Inhibitor Package
- Fibre Optic Cable
- Pig Launchers and Receivers
- Piping & tie-ins
- Foundation works
- Electrical equipment
- Control and Safeguarding systems
- Telecommunications equipment
- Pressure Vessels
- Cold Vent Stack
- Metering Package

# 4. Methodology

The methodology adopted in the review is a structured evaluation of different equipment packaging strategies and installation techniques to determine those to be recommended for adoption in the project based on criteria considered critical to a successful project execution. The criteria are defined in Section 4.1.3 of this document. The panel examined site installation activities and identified work-packs like piling, piping and cable works among others. The risks and constraints associated with each work-pack were identified along with recommendations to mitigate the risks. The assessment of the work-packs enabled the panel to generate a sequence of work that is considered appropriate for the project.

## 4.1. Evaluation Of Equipment Strategies And Installation Techniques

### 4.1.1. Equipment Strategy

Based on experience from previous projects, the following operational definitions were adopted for the equipment strategies identified and evaluated for adoption in the project:

#### 4.1.1.1.Skidding

Equipment strategy that involves yard fabrication and assembly of equipment on skids off-site. Each skid is brought to site with identified tie-in points for inter-connection with other skids.

#### 4.1.1.2.Modules

Equipment strategy that involves yard fabrication and assembly of all equipment on skids off-site into one or more modules. Each Module is brought to site with the component skids completely inter-connected with identified tie-in points with other modules.

#### 4.1.1.3.Piecemeal

This strategy involves the transportation of complete factory tested and fabricated vessels, assembled equipment, valves, flanged spool pieces, cable trays with electrical and instruments cable runs, and instruments to site in completed bits. Assembling and testing are then carried out on-site to form an integral unit, and the integral unit then hooked-up to identified tie-in points.

### 4.1.2. Installation Techniques

Based on experience from previous projects, the following operational definitions were adopted for the installation techniques identified. They were evaluated with respect to the criteria listed in Section 4.1.3 below to determine the most applicable which can be recommended for adoption:

#### 4.1.2.1Lifting

This installation technique involves loading equipment, skid or module on a barge or trailer for transportation to site and using a heavy lift crane on barge or wheels/track for positioning on previously prepared foundation structure.

##### 4.1.2.2Roll Over

An installation method whereby an equipment, skid or module is placed on powered rollers atop a transportation barge or trailer. At site, the barge or trailer is positioned adjacent to the previously prepared foundation structure and the skid or module, by means of the rollers is transferred from the barge into position on the foundation.

##### 4.1.2.3Float Over

This is an installation method that involves transporting a completely packaged module to site on a semi-submersible barge. At site, the barge is maneuvered carefully between previously prepared pile/jacketed structure and submerged when the mating surface of the module is vertically aligned with that of the pile/jacket. The submerging action lowers the module onto the pile/jacket. The barge is then retrieved in the submerged position.

#### 4.1.3. Strategy Review

In order to evaluate the identified equipment packaging strategies and installation techniques for applicability, the following criteria were identified as critical to the project and are defined below.

| CRITERIA | DEFINITION |
|---|---|
| Best Practices | Used in project of similar scope successfully in the last five years |
| Ease of Inter-facing | The potential of an equipment strategy or installation technique to minimize resources required for planning, engineering, procurement, field operations and site testing during the project. |
| Minimize HSE Risks | The potential of an equipment strategy or installation technique to minimize undue HSE risks |
| Ease of Access | The potential of an equipment strategy or installation technique to minimize logistics that will be required for personnel to get to fabrication yard or equipment to get to site.<br><br>The potential of a technique to minimize dredging required on the waterways or widening/strengthening of roads in order to transport men, materials and equipment to site. |
| Ease of meeting Project Schedule | This criterion is the potential of an option to facilitate the delivery of the project without delays beyond schedule |
| Ease of getting cost within budget | This criterion is the potential of an option to facilitate the delivery of the project without additional costs beyond budget. |

| | |
|---|---|
| Equipment Availability | The probability of sourcing the equipment required for installations locally. |

The equipment strategies and installation techniques that were identified were rated using the following defined scores.

| SCORE | DEFINITION |
|---|---|
| 1 | The strategy/technique being scored has low probability of achieving the criterion being considered. |
| 2 | The strategy/technique being scored has medium probability of achieving the criterion being considered. |
| 3 | The strategy/technique being scored has high probability of achieving the criterion being considered. |

The scores for each strategy or technique under various criteria were then summed up. Ranking was done in order of magnitude of the sum.

The strategy and technique with highest scores were considered to be most appropriate for the project.

### 4.1.4. Sequence Of Work

The sequence of work that may be adopted and developed for site work was identified and evaluated for suitability based on experience. Emphasis was placed on maximising off-site fabrication opportunities and utilizing open-cut with lay-barge and/or Horizontal Directional Drilling during project execution.

### 4.1.5. Workpack Review

Predetermined guidewords and catchphrases were used to review work-packs in line with common risks and constraints associated with the execution of different work-packs. The potential consequences are discussed, assessed and recorded along with actions for further consideration by the PMT. In reviewing each work-pack applicable to the project, the capability of local contracting companies to undertake the execution of the work pack was discussed.

The critical issues examined for identified work-packs are tabulated below.

| Work-Packs | Scope |
|---|---|
| Civil/Structural Works | Pile Foundations, Excavations, Removal of Debris. |
| Installations | Lifting/Rigging, Positioning, etc. |
| Onsite/Offsite Fabrication | Welding/Cutting/Joining, Lifting/Rigging/Towing, Scaffolding, Painting, etc. |
| Logistics/Transportation | Land/Marine/Aviation, Load Out, Offloading, Stacking, Storage, Retrieving, Delivery Schedule, etc. |
| Piping | Pipe Rack/Support Fabrication, Valve Positioning, Routing, etc. |
| Electrical/Instruments Cable Work | FOC, Cable Pulling, Cable Rack / Tray mounting, Terminations etc. |
| Tie-in Works | Drains, Relief, Utilities, Gas lines, Liquid Lines, etc. |
| Pipelines | Open-cut with lay-barge and/or Horizontal Directional Drilling. Lay-down & Pipe storage areas/barges, Access for boats for recovery of floaters/lowering-in, Trenching, Spoil heap accommodation, Pipeline welding access, NDT, coating, Cathodic Protection, cleaning/pigging, gauging, hydrotest, pre-commissioning, commissioning/start-up |

# 5. Findings

## 5.1 Equipment Strategy And Installation Techniques

Using the methodology described in section 4.1 of this document, the participants agreed as follows;

i. 'Skidding' equipment strategy is preferable for metering unit and pig traps
ii. 'Lifting' installation technique is favored over roll-over since all work locations are green fields with adequate space.
iii. Float-over is not applicable

## 5.2 Outcome Of Work Pack Review

The outcome of the exercise carried out in line with the procedure in 4.3 is recorded in Appendix 1 of this study. The major findings and specific recommendations are captured below.

- It was noted that the project will be executed with a single EPC contract. The scope needs to be comprehensive enough to include details that will ensure effective cost and HSE management.
- The contract will be given to an EPC contractor who will do the procurement of all materials.
- PMT to ensure that all key findings in LIRA study conducted for each location are managed to ALARP before construction.
- FEED Team shall ensure all discipline deliverables are harmonized before completion of FEED in order to avoid some discipline interface issues.
- ITT documentation should clearly address scope of utility requirement, HSE, Administration and SCD at construction site that the Contractor will be required to provide for own use during construction. The documentation should also include the responsible parties in order to clear all ambiguity.
- Contracting strategy should take cognizance of the fact that local companies have adequate capabilities to handle identified construction work-packs for the project.
- Involvement of Logistics department (particularly in PMT's LIRA documentation) before detail design close-out is highly required.
- Updating of existing security plan is necessary in order to capture recent security concerns. PMT shall ensure this is done
- The project design is predicated on using newly acquired ROW
- Where feasible, the locations already identified for installation of the various equipment packages shall be further optimized during detailed design
- There is need to probe and ascertain that areas to be impacted by foundation works and

excavations are free/not impacting on existing underground utilities.

- Given the sizes and weight of the Equipment, local availability of transportation trailers/barges and lifting devices are considered adequate. Notwithstanding, further optimization can further be done at detailed design.
- Provision of site services like lay-down area, temporary lighting, site offices/accommodation, site accommodation and security in view of multiple projects to be executed simultaneously could create Interface issues. Thus, the PMT should champion an interface meeting with the various stakeholders in order to resolve any interface issues that might arise during construction.

# 6. Recommendations

## 6.1. Construction Strategy

From the findings above, the construction strategy should involve fabrication, assembly and delivery of the various equipment to site in skid packages. Thereafter, simple lifting operation can be carried out using lift cranes to install and position the equipment. Appropriate lifting points on equipment should be addressed upfront prior to transportation and installation at site.

PMT should ensure that contractor makes provision for safety protection and safe working environment at site during construction and installation in the rainy season (such as lightning protection, etc.). This requirement should form part of the ITT package. The FEED Team shall be responsible for preparing the ITT package for the project.

The need to provide engineering data of the correct quality and at the required time to the equipment manufacturers, fabricators, vendors and constructors is fundamental to the success of the project approach. For this reason, the Vendor’s systems and procedures must be geared towards this, and all documents and materials should be coded and tracked accordingly. The contracting strategy to be fashioned should ensure single-point accountability for different packages from yard to installation and site testing. The prime vendor/contractor should be mandated to place emphasis on involving key sub-vendors in the early stages of the design to achieve maximum and optimized “Constructability”.

## 6.2. Work Sequence

The analysis of the project showed that on-site scope of the project involving equipment installation is best carried out in phases. The phases identified and sequences recommended are as follow: foundation works, Installation and positioning of the various equipment on the foundations and tie-in works.

The construction plan should accommodate underground works such as foundations of equipment, drainage systems, underground cablings, underground piping and roads early in the project to provide a clear workplace for subsequent installation of equipment and other above ground works.

## 6.3. Transportation

Project specific transportation study (LIRA) should be carried out to address the issues pertaining to air, land and marine transportation.

Review of availability of jetty facilities should be captured in the Transportation Study (LIRA). Adequacy of river transportation routing with respect to water depth / dredging, EIA, and local regulations should be properly documented.

The specific techniques/procedure of handling the line-pipes and other equipment, both at yard load-out, during transportation and installation at site should be agreed before placement of order. There are few contractors capable of undertaking integrated transportation and installation contracts and the selection of their particular mobile equipment can have an influence upon the project cost and schedule. An important element for early consideration is the availability of watercraft that can handle lifting of heavy equipment with own gears, and vessels that can transport heavy equipment.

If a local company will be engaged in the coating of line pipes and fabrication of any of the packages and load-out will involve road transportation, the package design should take cognizance of dimensional and weight limitations for road transport in the country, which stipulates maximum tonnage of 45Ton and dimension of Package of 9 x 3.7 x 4.6 m (length x breadth x height) for packages. Issues pertaining to stress changes in package frames during lifting and transportation should be addressed in the package design.

In line with Nigerian Customs law, items being imported into the country are liable to customs clearance. Project execution plan shall take cognizance of the place, time and resources required for such clearing activities.

## 6.4. Local Content

In view of Federal Government's Nigerian Content Policy, the capabilities of local companies with regards to the scope of on-site works, structural/equipment fabrications, pipe-work and logistics should be considered. The project contracting strategy should take advantage of this capability accordingly. Accordingly, it will be required to specify the percentage of local content in the ITT in such a way that the EPC contractor will have the responsibility of engaging local vendors where feasible.

## 6.5. Pipelines

- The Contractor should ensure that the impact of movement of heavy equipment across Pipeline & Flowline corridors is addressed and documented prior to mobilization to site.
- The workshop agreed two (2) main methods of pipeline construction; open-cut using lay-barge with HDD for river crossings and continuous HDD. These methods are to be included in the ITT.
- Further reference should be made to the attached constructability worksheet (Attachment 1 of Section 8.0)

## 6.6. Others

- Sand filling should begin after ascertaining non-existence of underground facilities in the impacted areas. This will ensure consolidation prior to the main construction works.
- The weights of the various equipment should be confirmed during detailed design. This is to ensure crane capacity estimation and transportation requirements are properly addressed before and during contracting.
- Interface issues that could arise in the provision of site service like temporary lighting, site offices, site accommodation and security should be critically analyzed and planned out before site mobilization.
- Assess existing sources of water for adequacy during construction.
- OR & A Team should be involved and adequately represented throughout the Project duration; and there should be an agreed plan towards ensuring flawless project delivery, start-up and operations.

# 7. Conclusion

The identified work phases and recommended construction strategy and sequence will help meet not only the key project requirements of minimum site exposure, ease of installation and cost effectiveness but also enable HSE risks to be minimized. Furthermore, it will be in compliance with the Nigerian Content Policy for the fabrications to be carried out in Nigeria wherever such is feasible.

# 8. Atachments

## 8.1. ATTACHMENT 1: Constructability Study Worksheet

Constructability Study Worksheet

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **CONSTRUCTABILITY REVIEW WORKSHOP WORKSHEET** | | | | | | | | | |
| | | **PROJECT** | **FEED for Gas Gathering Pipelines & Associated Infrastructure** | | | | | | | | |
| | | **Workshop Date** | **3rd July, 2022** | | **NODE(S) / LOCATION(S)** | **Boubouwe Bou (BOU), Elepa (ELEP), Akabel (AKAB)** | | | | | |
| | | **CONSTRUCTION ASPECT** | | **GENERAL** | **MAJOR ITEMS / EQUIPMENT / MATERIALS** | **Lay-barge, Horizontal Directional Drilling rig, drill pipes/bits, mud-tanks, piling rig, casing pipes, corrosion inhibitor package, pressure vessels, RTU, and Electrical equipment** | | | | | |
| | | | | **COMMENTS** | | | | | | | |
| **Row No** | **ISSUES** | **CRITERIA** | **Pertinent Questions / Concerns** | **REFFERENCES** | **RESPONSE, CONSTRAINTS AND RISKS** | **IMPACTS** | **RECOMMENDATION / ACTIONS TO BE TAKEN** | **RANKING OF RECOMMENDATION H-Important/Critical/Standard/Must do ; M-Suggestion for Construction Optimization (e.g. schedule/cost reduction) ; L-Documentation Improvement ; N/A - Not Applicable** | **RESPONSIBLE PARTY** | **TIMELINE** | **OTHER REMARKS** |
| **1** | Equipment Packaging and Layout Considerations | Equipment Packaging Strategy/Equipment List | Has a Philosophy and or Equipment List been generated to showcase the Strategy for packaging equipments on the project (Skids, Modules etc)? | | There is an Equpment List for electrical & instrumentation. There is line list. Process equipment list is under preparation. Skidding units will be suitable for the equipments. | HSE, Schedule, Cost, Construction & Installation. | Ensure that the FEED highlights the suitability of skidding of metering units, pig traps and chemical injection package. | M | FEED Team | Before Finalisation of FEED Design. | |
| **2** | | Identification | Which equipment package can be categorized as major, for further constructability studies at this time (size, weight, complexity as main criteria) | NONE | Pig traps, Corrosion Inhibitor Package, Closed Drain Vessel, Open Drain Tank, Cold Vent stack, RTU and Electrical equipment are the major equipment; but since there is no complexity of form, size or weight, the packages can be safely handled with regular handling /transportation equipment. | HSE, Schedule, Cost, Construction & Installation. | No special studies will be required. | M | N/A | N/A | |
| **3** | | Package Limit | Are the package limits clearly delineated? | See PFS and PEFS | No. It will be shown in the PEFS | HSE, Schedule, Cost, Construction & Installation. | Confirm that all Package limits are clearly shown in drawings. | H | FEED Team | Before finalising the design | To look at the PEFs |
| **4** | | Best practices | Are the package limits appropriate / in line with Global Best practices? | PEFS and specification documents | No. To be finalised after item 3 above. | HSE, Schedule, Cost, Construction & Installation. | As in item 3 above.M | H | FEED Team | Before finalising the design | Any special tools requirements for any equipment to be installed should be identified upfront with Vendor(s) and provided. This should form part of the ITT. |
| **5** | | Specification of Packages | Do the Specifications generated for the Packages capture the applicable Package Limit for each Package? | Specification documents | Yes for the specification documents that have been produced so far. . | HSE, Schedule, Cost, Construction & Installation. | Ensure that the remaining Specification documents capture the applicable package limits. | M | FEED Team | Before FEED completion | |
| **6** | | Size of Packages | Have the footprints of the packages been estimated? If yes, which one is the largest? | Reference Equip list | Yes, for the pig launchers and receivers ; No, for all other packages. But they are all items that can be readily transported on the roads. | HSE, Schedule, Cost, Construction & Installation. | None | M | N/A | N/A | |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **CONSTRUCTABILITY REVIEW WORKSHOP WORKSHEET** | | | | | | | | | |
| | | **PROJECT** | **FEED for Gas Gathering Pipelines & Associated Infrastructure** | | | | | | | | |
| | | **Workshop Date** | **3rd July, 2022** | | **NODE(S) / LOCATION(S)** | **Boubouwe Bou (BOU), Elepa (ELEP), Akabel (AKAB)** | | | | | |
| | | **CONSTRUCTION ASPECT** | | **GENERAL** | **MAJOR ITEMS / EQUIPMENT / MATERIALS** | **Lay-barge, Horizontal Directional Drilling rig, drill pipes/bits, mud-tanks, piling rig, casing pipes, corrosion inhibitor package, pressure vessels, RTU, and Electrical equipment** | | | | | |
| | | | | **COMMENTS** | | | | | | | |
| **Row No** | **ISSUES** | **CRITERIA** | **Pertinent Questions / Concerns** | **REFFERENCES** | **RESPONSE, CONSTRAINTS AND RISKS** | **IMPACTS** | **RECOMMENDATION / ACTIONS TO BE TAKEN** | **RANKING OF RECOMMENDATION H-Important/Critical/Standard/Must do ; M-Suggestion for Construction Optimization (e.g. schedule/cost reduction) ; L-Documentation Improvement ; N/A - Not Applicable** | **RESPONSIBLE PARTY** | **TIMELINE** | **OTHER REMARKS** |
| **7** | Equipment Packaging and Layout Considerations | Weight of Packages | Have the weights of the packages been estimated? If yes, which one is the heaviest? | Reference Equip list | No | HSE, Schedule, Cost, Logistics, Construction & Installation. | Weights to be known during DED | M | PMT/FEED team | During DED | |
| **8** | | Layout | Is the Layout sufficiently frozen for the scope of Work? | Layout Drawings | YES | Schedule, Cost, Construction & Installation. | NO FURTHER DISCUSSION REQUIRED | M | PMT/FEED team | Share the drwaings | |
| **9** | | | Which equipment packaging strategy is best suited to Layout ( Modules / Skidding / Piecemeal)? | See study document | Skidding | Schedule, Cost, Construction & Installation. | None | M | N/A | N/A | Packaging strategy to be re- assessed in case further update of layout makes that necessary. |
| **10** | | | Are piecemeal / skid / module integrated sizes suited to location? | Layout Drawings | No. Refer to item 8 above. | No. Refer to item 8 above. | No. Refer to item 8 above. | No. Refer to item 8 above. | No. Refer to item 8 above. | No. Refer to item 8 above. | |
| **11** | | | Are cranes with adequate tonnage available to lift the skids/modules? Else, what lifting strategy will be employed? | | NO, Refer to item 7 above | Schedule, Cost, Construction & Installation. | Refer to item 7 above | M | CONTRACTOR | BEFORE Site works | |
| **12** | | Capability List | Is there a list of capabilities that a vendor must have in order to be engaged for specific scope of work in this project ? | PMT | Yes, Pre-qualification criteria exist, and EPC Tendering is ongoing amongst already pre-qualified Contractors | N/A | None | L | N/A | N/A | TO BE SHARED |
| **13** | | Nominations | Have vendors been nominated for major scopes and checked against the required capabilities? | PMT | YES, In line with prequalification criteria | HSE, Schedule, Cost, Logistics, Procurement, Construction & Installation. | Carry out sub-contractor and sub-vendor nomination for major scopes and check vendors against the required capabilities | M | PMT | Before ITT | |
| **14** | | Nigerian Content | Are there local vendors/fabricators that can be engaged for procurement? Name them. | | Yes, names to be confirmed | HSE, Schedule, Cost, Logistics, Procurement, Construction & Installation. | Confirm and incorporate relevant requirement in the ITT in line with NCD requirements. | H | PMT/FEED team | Before ITT | |
| **15** | | | Do the vendors need further development prior to engagement? | | No, Vendors who do not meet requirements will not be selected. | N/A | None | None | N/A | N/A | |
| **16** | | | Is contracting strategy agreed? If no, when will it be? What kind of strategy is it? | | Yes: It is a single EPC ( engineering, procurement, construction) | N/A | None | None | N/A | N/A | |

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| **17** | | Contracting Strategy | Will there be a robust handshake between design and (likely) strategy(ies)? | | Yes. Same Contractor will carry out detailed design, procurement of all materials and construction. With this strategy, interface issues are expected to be minimal. | HSE, Schedule, Cost, Logistics, Procurement, Construction & Installation. | EPC Contractor to review, accept and own FEED at start of EPC contract. FEED Contractor to provide support as may be required. | H | PMT | Before Order Placement | |
| **18** | Vendor / Procurement / Fabrication | Long Lead Items | Are long lead items and critical items identified? Are plans in place to deliver them at the right time and cost? | | Yes , (manifold, pig launchers/ receivers, pipeline, fibre optic cables) long lead items have been identified. Yes, project schedule is in place and cost control will be applied. | Schedule, Cost, Construction & Installation. | Apply cost control to the procurement of the long lead items. | M | PMT | Before ITT | |
| **19** | | | Will specific equipment package/material delivery sequence be required? What impact may that have? | | Yes | Schedule, Cost, Construction & Installation. | Delivery Sequence : The linepipes to come and be coated before construction, then foundation & corrosion materials; other heavy equipment within project scope to be delivered. | M | PMT | Before ITT | |
| **20** | | Sequence | Any recommendation on sequence of fabrication work? | | Yes | HSE, Schedule, Cost, Logistics, Procurement, Construction & Installation. | Engage vendor and ensure that long lead items are taken on-board first in line with Project Plan. Also refer to item 19 above. | H | PMT | Before Order Placement | |
| **21** | | Schedule | What is the proven lead time for procurement of major packages? | | 12 months plus | Schedule, Cost, Logistics, Procurement, Construction & | None | None | N/A | | |
| **22** | | | What is the proven transportation time/cost for packages(say, from Europe)? | | At least 4 weeks | Schedule, Cost, Logistics, Procurement, Construction & Installation | None | None | N/A | N/A | |
| **23** | | | What is the proven clearing/expediting time for packages in Nigeria? Is there any favourable initiative or waiver/exemption? | | About 2 weeks. IDEC to be extended to current project where feasible to do so. | Schedule | Confirm if IDEC can be extended to current project. | M | PMT | Before ITT | |
| **24** | | Materials | Are local bulk materials supply applicable/available? | | Bulk materials (pipes, valves, etc.) are applicable.However, availability needs to be confirmed | HSE, Schedule, Cost, Logistics, Procurement, Construction & Installation. | Confirm availability of bulk materials and if not available, consider alternative procurement strategy. | M | | Before finalising contracting strategy. | |

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| **25** | | Transport Studies | Has a transportation study been conducted for the project? If yes, what are the key findings? | | In progress. | Logistics | Complete ongoing Logistic Infrastructure and Resource Assessment (LIRA) for the project to address the identified issues and associated capabilities of available transportation infrastructure. | H | PMT | During FEED | |
| **26** | Logistics/Transport | Weight/Size | In line with the package limits discussed above, is there any weight or size restriction? If yes, how will it be resolved? | | No restriction. Weight and size of the pig launcher can be transported on Nigerian roads /water ways. | | None | None | N/A | N/A | |
| **27** | | Infrastructure | Is necessary infrastructure in place (jetties / helipads, road, parking lot, etc.) ? Is it adequate? | | There are existing roads, water ways. LIRA study to look at the various aspects. | Logistics / Procurement / Schedule, Cost. | Carry out LIRA to address the identified issues and associated capabilities of available transportation infrastructure. | H | PMT | Before ITT | There are associated security and safety concerns regarding transportation. The assessment of these concerns should be captured in the LIRA study. |
| **28** | | Weather | Will transportation require any weather window (Season/Tide, etc.)? | | Refer to LIRA | Logistics / Procurement / Schedule, Cost. | Carry out LIRA to address the identified issues. | H | PMT | Before ITT | |
| **29** | Logistics/Transport | Mode(s) | What mode (s) of transportations will be employed( land, marine, aviation)? What issues are likely to arise? And how will they be mitigated? | | Land , Marine, Aviation . LIRA to capture issues (such as security, availability of infrastructure) | Logistics / Procurement / Schedule, Cost. | Carry out LIRA to address the identified issues and associated capabilities of available transportation infrastructure. | H | PMT | Before ITT | |
| **30** | Temporary Storage | Laydown Area | Are there enough laydown areas for construction equipment and materials? | | The construction contractor will store in own facility or coating plant prior to transportation to site. A sandfilled area for temporary | Schedule, Cost, Construction. | To confirm availablility/requirement before ITT. | H | PMT | Before ITT | |
| **31** | | Construction Method | What contruction method(s) will be employed for the pipelines? | | 1. Open-cut using lay-barge with HDD for river crossings<br>2. Continous HDD. | Logistics / Procurement / Schedule, Cost. | Address the construction methods to be adopted during pipelines construction and installation, assess possible impact on settlements. Note required actions, and capture in ITT. | H | PMT | Before ITT | |
| **32** | | Equipment | What (specialised) equipment will be required? And are they readily available? | | 1. Drilling rig<br>2. Drill-hole tracking system<br>3. Mud-tank<br>4. Lay-barge<br>YES, readily available | Logistics / Procurement / Schedule, Cost. | Itemise all specialised equipment to be used during pipelines construction and installation, confirm availability. Note required standards and actions, and capture in ITT. | H | PMT | Before ITT | |
| **33** | | Routing / ROW | Is existing ROW to be used or a new acquisition is required? | | New RoW being **partially** acquired | Logistics / Procurement / Schedule, Cost. | Firm up required actions and capture in ITT. | H | PMT | Before ITT | |
| **34** | Pipelines | Routing / Vegetation | Any crossing of tropical scrubland / vegitation / undergrowth? How will spread width be addressed? | | Mangrove vegetation (re-vegetation to be done on open-cut areas) | Logistics / Procurement / Schedule, Cost. | Ascertain activities relating to crossing of tropical scrubland & undergrowth , firm up required actions and capture in ITT. | H | PMT | Before ITT | |

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| **35** | | Routing / Swamp | Will there be crossing of swampy areas and/or areas subject to flooding? How will buoyancy issue be addressed? If required, how will the water level be maintained? | | YES | Logistics / Procurement / Schedule, Cost. | Line pipes will be concrete coated to provide the required negative bouyancy for open-cut sections | H | PMT | Before ITT | | |
| **36** | | Routing / Rivers | Any river crossings (major, medium, minor)? What technique(s) will be considered for wide & deep crossings? Is the river fast flowing? Is scouring / erosion possible? How will buoyancy issue be addressed? If required, how will the water level be maintained? | | YES (Major crossings) | Logistics / Procurement / Schedule, Cost. | 3 river crossing having 1-3km width Technique; HDD recommended Ascertain activities relating to erosion/width/depth/buoyancy for river crossing , firm up required actions and capture in ITT. | H | PMT | Before ITT | | |

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| **37** | Pipelines | Routing / Roads | Any road crossings (major, minor)? | | NO | Logistics / Procurement / Schedule, Cost. | Ascertain the location of proposed Nembe-Brass road relative to the pipelines and work activity required and capture in ITT. | H | PMT | Before ITT | |
| **38** | | Routing / existing pipelines | Any pipeline crossings? | | YES<br>4pipelines (3 owned by NAOC & 1 owned by SPDC) | Logistics / Procurement / Schedule, Cost. | HDD is recommended for the crossings<br>Ascertain the diamters/depth of the pipeline crossings, access the impacts and work activity required | H | PMT | Before ITT | |
| **39** | | Interdiscipline Issues | Are the various interfaces already taken care of (Disciplines, etc.)? | | In progress. | Schedule, Cost, Construction & Installation. | All disciplines to harmonise design documentation | H | FEED Team | Before finalisation of Design. | |
| **40** | Installations | Inter-functional Issues | Are the various interfaces already taken care of (PMT, Construction, Vendor, etc.)? | | In progress. | HSE, Schedule, Cost, Logistics, Procurement, Construction & Installation. | Continue ongoing interfaace meetings and engagements between the various stakeholders (SPDC, Government reulators and authorities, Communities, etc.) to address interface issues.<br>Continue ongoing engagements with the responsible authorities for the required permits and consents.<br>Finalise an interface register with action parties and timelines. | H | PMT | Before EPC Contract award | |
| **41** | | Installation Sequence | Is there any specific installation sequence to follow? Identify the sequence. | | Yes. | Cost, Schedule. | EPC contractor to employ multiple spreads.<br>Cased pile for manifold platform | H | PMT | During Construction & Installation | |
| **42** | Commisioning | Pre-Commisioning | Availability of Air, Water, Utilitiesetc. | | Contractor provides the air, portable water, electricty, nitrogen (utilities) etc. | HSE, Schedule, Cost | Put this in the ITT | M | PMT | Before ITT | Contractor to provide all utilities required for construction work. |
| **43** | | Requirements | Gas availability | | Yes, Gas is available. | HSE, Schedule, Cost | Gas will be available from SPDC | M | PMT | Before ITT | |
| **44** | | | Is planned shutdown required? How long? | | Not required | HSE, Schedule, Cost, Construction & Installation. | N/A | N/A | PMT | | |
| **45** | | | What systems should be shut down? For how long? | | Not required | HSE, Schedule, Cost, Construction & Installation. | N/A | N/A | PMT | | |
| **46** | Shutdown | Requirements | What are the activites to be done during shut downs? | | N/A | HSE, Schedule, Cost, Construction & Installation. | N/A | N/A | PMT | | |
| **47** | | | Can the shut down be avoided? How can it be minimised? | | N/A | HSE, Schedule, Cost, Construction & Installation. | N/A | N/A | PMT | | |
| **48** | | | Cost of avoidance | | N/A | Cost, Schedule. | N/A | N/A | PMT | | |
| **49** | | | How will Power Supply be arranged for construction? | Contracting Strategy and Construction Plan. | Contractor to provide power | HSE, Cost, Construction & Installation. | Capture in the ITT | M | PMT | Before ITT | |

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| 50 | Construction Site | Utility Requirements | How will Water Supply be arranged for construction? | Contracting Strategy and Construction Plan. | Contractor to provide water supply | HSE, Cost, Construction & Installation. | Capture in the ITT | M | PMT | Before ITT | |
| 51 | | | How will Drainage and Waste collection/treatment be arranged for construction? | Contracting Strategy and Construction Plan. | Contractor to provide functional waste mgt system statutory requirements | HSE, Cost, Construction & Installation. | Capture in the ITT | M | PMT | Before ITT | |
| 52 | | | Where will Site Office be located? What size? Is there space for it? | Contracting Strategy and Construction Plan. | Contractor to provide site office. | HSE, Cost, Construction & Installation. | Capture in the ITT | M | PMT | Before ITT | |
| 53 | Construction Site | Utility Requirements | Security | Contracting Strategy and Construction Plan. | Contractor to provide | HSE, Cost, Construction & Installation. | Capture in the ITT | M | PMT | Before ITT | |
| 54 | | | Construction Accommodation Camp | Contracting Strategy and Construction Plan. | Contractor to provide | HSE, Cost, Construction & Installation. | Capture in the ITT | M | PMT | Before ITT | |
| 55 | | | Temporary Lighting | Contracting Strategy and Construction | Contractor to provide | HSE, Cost, Construction & | Capture in the ITT | M | PMT | Before ITT | |
| 56 | | HSE | HSE | Contracting Strategy and Construction Plan. | Contractor to provide in line with Client policy | HSE, Cost, Construction & Installation. | Capture in the ITT | M | PMT | Before ITT | |
| 57 | | Administration | Construction Organogram | Contracting Strategy and Construction Plan. | Contractor to provide. PMT to provide for execution team. | HSE, Cost, Construction & Installation. | Capture in the ITT | M | PMT | Before ITT | |
| 58 | | SCD | SCD | Contracting Strategy and Construction Plan. | Client to lead Combination of Client & Contractor. Client or Contractor to secure FTO Contractor to develop and implement an SCD plan | HSE, Cost, Construction & Installation. | CSR Plan being developed BY BFPCL Capture in the ITT | M | PMT | Before ITT | |

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| **59** | Employment | Employment | Employment of locals at construction stage of project | Contracting Strategy and Construction Plan. | Contractor to implement in line with the FTO | HSE, Cost, Construction & Installation. | The GMOU needs to be put in the ITT | M | PMT | Before ITT | |
| **60** | Security | Security | Security concerns | Project Plan | BFPCL & Contractor to handle | HSE, Cost, Construction & Installation. | Sglobal security outfit set up by BFPCL<br>Security plan to be prepared by PMT | M | PMT | Before ITT | |
| **61** | OR & A | OR & A | Is there an existing OR & A plan for the project? | Project Plan | Yes. Operations and Maintenance Philosophy developed, and Performance Guarantee & Test Procedure inplace (includes OR & A) | HSE, Schedule, Cost, Construction & Installation. | To be issued with ITT | M | PMT | Before ITT | There is OR & A engineer to follow up and ensure project compliance with requirements. |